

量子统计方法

1. 密度矩阵

以 $|\psi_i\rangle$ 表征量子系统的某态, 其出现概率为 P_i : 若系统可表示为某态 $|\psi\rangle$, $P_i = |\langle\psi|\psi_i\rangle|^2 = c_i^* c_i$ 纯态, $\sum P_i = 1$
不可表示, ψ_i 对应 P_i 混合态

密度矩阵 $\hat{\rho} = \sum |\psi_i\rangle P_i \langle\psi_i| \Rightarrow \text{tr}(\hat{\rho}) = \sum P_i = 1$ (纯态 $\hat{\rho} = |\psi\rangle\langle\psi|$)

力学量期望: (纯态) $\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle = \sum c_i^* c_i \langle\psi|\hat{A}|\psi_i\rangle + \sum_{i \neq j} c_i^* c_j \langle\psi|\hat{A}|\psi_j\rangle$

$\uparrow$ 纯态时 $\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle \cdot 1$, 即 $\langle\hat{A}\rangle = 1$

(混合态) $\langle\hat{A}\rangle = \sum P_i \langle\psi_i|\hat{A}|\psi_i\rangle = \sum P_i A_i$, $A_i = \langle\psi_i|\hat{A}|\psi_i\rangle$ 为 ψ_i 态的平均

ϕ_n 为 $\hat{A}$ 本征矢, $\sum |\phi_n\rangle\langle\phi_n| = 1$

$$1) \text{tr}(\hat{A}\hat{\rho}) = \langle\hat{A}\rangle \quad \langle\hat{A}\rangle = \sum_i P_i \langle\psi_i|\hat{A}|\psi_i\rangle = \sum_n \sum_i P_i \langle\psi_i|\phi_n\rangle \langle\phi_n|\hat{A}|\psi_i\rangle = \sum_n \langle\phi_n|\hat{A}|\underbrace{\sum_i \langle\psi_i|\psi_i\rangle P_i}_{\hat{\rho}}|\phi_n\rangle$$

★ $= \text{tr}(\hat{A}\hat{\rho})$ $\text{tr}(\hat{\rho}) = 1$

$$2) \hat{\rho}|\psi_i\rangle = P_i|\psi_i\rangle \quad \text{对某态 } \psi_i, \text{ 若 } \psi_i, \psi_j \text{ 正交, 则有: } \hat{\rho}|\psi_i\rangle = \sum |\psi_j\rangle P_j \langle\psi_j|\psi_i\rangle = P_i|\psi_i\rangle$$

纯态: $\hat{\rho} = |\psi\rangle\langle\psi| \Rightarrow \hat{\rho}|\psi\rangle = \langle\psi|\psi\rangle|\psi\rangle$

$$3) \text{tr}\hat{\rho} = 1 \quad \text{纯态: } \text{tr}\hat{\rho} = \text{tr}\hat{\rho} = 1$$

混合态: $\text{tr}\hat{\rho} < 1$

$$4) \hat{\rho}^\dagger = \hat{\rho} \quad \text{pf: } \rho_{nm} = \langle\psi_n|\hat{\rho}|\psi_m\rangle = \sum \langle\psi_n|\psi_i\rangle P_i \langle\psi_i|\psi_m\rangle$$

$$\rho_{nm}^* = \langle\psi_m|\hat{\rho}|\psi_n\rangle = \sum \langle\psi_i|\psi_m\rangle P_i \langle\psi_i|\psi_n\rangle$$

] 等, done

量子刘维尔方程/冯诺依曼方程:

$$\hat{\rho}(t) = \sum |\psi_i(t)\rangle P_i \langle\psi_i(t)| \Rightarrow i\hbar \frac{\partial}{\partial t} \hat{\rho} = i\hbar \frac{\partial}{\partial t} |\psi_i\rangle \langle\psi_i| \cdot P_i + |\psi_i\rangle \langle\psi_i| \cdot i\hbar \frac{\partial}{\partial t} P_i$$

$$= \hat{H} \sum |\psi_i\rangle \langle\psi_i| P_i - \sum |\psi_i\rangle \langle\psi_i| P_i \hat{H} = [\hat{H}, \hat{\rho}]$$

$i\hbar \frac{\partial}{\partial t} \hat{\rho} = [\hat{H}, \hat{\rho}]$ (该方法可用于H表象的相空间展开)

稳定时 $\frac{\partial}{\partial t} \hat{\rho} = 0 \Rightarrow [\hat{H}, \hat{\rho}] = 0$ ★

2. 量子系综表示:

统计假设: 等概率: $P_i = \frac{\Omega_i}{\Omega(E)}$ $\Rightarrow$ 密度矩阵 $\hat{\rho} = \sum |\psi_n\rangle P_n \langle \psi_n| = \sum |\psi_n\rangle \frac{\Omega_n}{\Omega} \langle \psi_n|$, $|\psi_n\rangle$ 为 $\hat{H}$ 本征

$$\text{tr} \hat{\rho} = \frac{\sum \Omega_n}{\Omega} = 1$$

$$\text{能定时: } \text{tr} \hat{H} \hat{\rho} = \text{tr} \hat{\rho} \hat{H} = \frac{\sum H_n \Omega_n}{\Omega} = \langle E \rangle$$

$$\text{定义 } \hat{S} = -k_B \ln \hat{\rho} \Rightarrow \langle \hat{S} \rangle = -k_B \text{tr}(\hat{\rho} \ln \hat{\rho}) = -k_B \sum P_n \ln P_n$$

微正则系: N 个孤立系统

相邻系统不作用: $P_{mn} = P_n \delta_{mn}$

$$P_n = \frac{1}{\Omega(E)}, \Omega(E) \text{ 为 } [E, E+\Delta E] \text{ 的微观数(总态数)}$$

$$\hat{\rho} = \sum |\psi_n\rangle \langle \psi_n| \cdot \frac{1}{\Omega(E)} = \frac{1}{\Omega(E)} \text{ (等概率)}$$

$$\langle S \rangle = -k_B \text{tr}(\hat{\rho} \ln \hat{\rho}) = -k_B \sum \frac{1}{\Omega(E)} \ln \frac{1}{\Omega(E)} = k_B \ln \Omega(E)$$

正则系: 能量交换, 粒子数不交换

(用配分函数推导) 配分函数有: $P_i = \frac{1}{Z} e^{-\beta E_i}$

$$\hat{\rho} = \sum |\psi_i\rangle \frac{1}{Z} e^{-\beta E_i} \langle \psi_i| = \sum \frac{1}{Z} e^{-\beta E_i} |\psi_i\rangle \langle \psi_i| = \frac{1}{Z} e^{-\beta \hat{H}} \cdot \frac{\sum 1 \times 1}{1}$$

$$Z = \sum e^{-\beta E_i} = \text{tr}(e^{-\beta \hat{H}}) \text{ (归一化常数)} = \frac{1}{Z} e^{-\beta \hat{H}}$$

$$\text{若 } \partial \hat{A} = 0 \Rightarrow \langle A \rangle = \text{tr}(\hat{A} \hat{\rho}) = \text{tr}(\hat{\rho} \hat{A}) = \frac{\text{tr}(\hat{A} e^{-\beta \hat{H}})}{\text{tr}(e^{-\beta \hat{H}})}$$

(用系综相互作用推导) 考虑外界: 内 $\otimes$ 外: $|\Phi_n\rangle = \sum_i |\psi_i\rangle |\varphi_n\rangle c_{in}$

$$\hat{\rho}_0 = \sum |\Phi_n\rangle P_n \langle \Phi_n| = \sum_n \sum_{i,j} |\psi_i\rangle |\varphi_n\rangle \langle \psi_j| \langle \varphi_n| P_n c_{in} c_{jn}^* \text{ , } P_n = \frac{1}{\Omega(E)}$$

$$\hat{\rho} = \text{tr}_{\text{外}} \hat{\rho}_0 = \sum_n \langle \varphi_n | \hat{\rho}_0 | \varphi_n \rangle = \frac{1}{\Omega(E)} \sum_n \sum_{i,j} \sum_{\alpha} \delta_{\alpha\alpha'} |\psi_i\rangle c_{in} c_{jn}^* \langle \psi_j| \delta_{\alpha\alpha'}$$

$$= \frac{1}{\Omega(E)} \sum_n \sum_{i,j} |\psi_i\rangle c_{in}^* c_{jn} \langle \psi_j| \text{ (外界相互作用孤立, 内+外也相互作用孤立)}$$

$$\text{代入: } c_{in} = \langle \varphi_n | \psi_i \rangle \text{ , 有: } \hat{\rho} = \frac{1}{\Omega(E)} \sum_n \sum_i |\psi_i\rangle \langle \varphi_n | \langle \varphi_n | \sum_j |\varphi_n\rangle \langle \varphi_n| |\varphi_n\rangle \langle \varphi_n|$$

$$= \frac{1}{\Omega(E)} \sum |\psi_i\rangle \langle \varphi_n | \langle \varphi_n | \varphi_n \rangle \langle \varphi_n |$$

$$= \frac{1}{\Omega(E)} \sum |\psi_i\rangle \sum_n \langle \varphi_n | \varphi_n \rangle \langle \varphi_n |$$

$$= \sum |\psi_i\rangle \frac{\Omega(E - E_i)}{\Omega(E)} \langle \varphi_n | \Rightarrow P_i = \frac{\Omega(E - E_i)}{\Omega(E)} = e^{-\beta(E - E_i)} = e^{-\beta E_i} \text{ (} E \gg E_i \text{, 近似)}$$

$$\text{对 } P_i \text{ 展开: } \frac{1}{k_B} [S(E - E_i)] \approx \frac{S(E)}{k_B} - \beta E_i \text{ , } k_B \beta = \frac{\partial S}{\partial E_i} = \frac{1}{T} \Rightarrow P_i = e^{-\beta E_i} \text{ (} E \gg E_i \text{, 近似)}$$

热力学函数 自由能 $F = -k_B T \ln Z$

巨正则系: 交换能量, 粒子数

配分函数:

$$P_{n,m} = \frac{1}{\Xi} e^{-\beta(E_n - \mu N_n)} \Rightarrow \hat{\rho} = \sum_{n,m} |\psi_n\rangle P_{n,m} \langle \psi_n| = \frac{1}{\Xi} e^{-\beta(\hat{H} - \mu \hat{N})}$$

$$\Xi = \sum_{n,m} e^{-\beta(E_n - \mu N_n)} = \text{tr} e^{-\beta(\hat{H} - \mu \hat{N})}$$

$$= \sum_n z^n Z_N(T, V)$$

$$\text{热力学势 } \Omega = -\frac{1}{\beta} \ln \Xi(T, V, \mu)$$